CODENEXUS / NSEC / TEAM CHROLLO

Dante

Brand change and drift tracking

A source-linked review workspace for teams that need every claim to stay current.

The problem Dante solves

Product facts, brand rules, approved terms, and visual assets often live in separate files and channels. New content can repeat outdated claims, while older material stays online after the underlying fact changes. Teams need to find affected items, see evidence, and route decisions to a person.

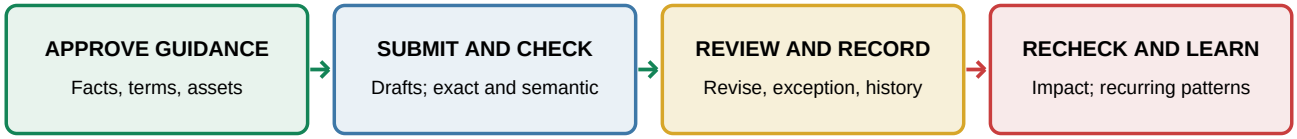
Why this problem matters

Lucidpress, *The State of Brand Consistency* (2019), surveyed 200+ organizations. Historical, self-reported results; the two questions are distinct and are not a current benchmark. Open report.

Who it serves

TEAM	NEED
Brand and marketing	Keep approved claims consistent across campaigns and channels.
Creators and designers	Receive specific evidence and a useful correction path.
Brand reviewers	Assess evidence, decide exceptions, and track changes.
Product and communications	Update claims after facts or policies change.
Distributed teams	Share current guidance with employees and agencies.

How the product works



Full sequence: add guidance, approve its version, submit text and channel, check evidence, revise or record an exception, save the decision, recheck linked content after source changes, and review recurring patterns.

Demo scenario: an approved storage claim changes from 100 GB to 200 GB. Dante identifies linked drafts that still say 100 GB, shows the source version and evidence, and sends each item to a reviewer.

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Major features Dante offers

A working workspace for governed content review, with clear boundaries between automated checks and human decisions.

Approved source library

Create facts, rules, required terms, and logo assets. Keep approved versions and history.

Evidence-linked content review

Run exact quantity and term checks. Optional Gemini rule review returns source and draft quotes for a person to assess.

Change impact analysis

Preview linked content affected by a source update; stale or conflicting content returns to review.

Human review and exceptions

Reviewers revise content or record a reasoned exception. Source changes and decisions enter the audit trail.

PDF and DOCX policy intake

Extract likely rules with page or paragraph references. Administrators approve candidates before they become active.

Scanned PDF OCR

When a page has little selectable text, the uploader can opt in to Gemini OCR. Candidates remain unapproved and carry a verification warning.

Campaign image review

With explicit consent, Gemini transcribes visible copy and gives advisory logo and contrast findings; an approved logo image can be compared.

Public page monitor

Capture and check public, server-readable page text on demand; reviewers decide whether a change affects guidance.

Drift analytics

Recharts show saved review activity by time, channel, campaign, owner, and current status. These are workspace signals, not accuracy scores.

Workspace access and audit

Supabase Auth, workspace roles, email-bound invitations, private files, and CSV/JSON audit export.

Publishing gate

A reviewer authorizes current content before Dante sends it to a configured generic HTTPS webhook.

Safety controls

Server-side authorization, source-version checks, opt-in Gemini calls, private storage, and reset protection for persistent workspaces.

Dante does not auto-approve extracted rules or publish content without a reviewer action. Gemini outputs remain advisory.

Round 3 focus areas and current status

The three planned areas are listed below with an honest implementation status.

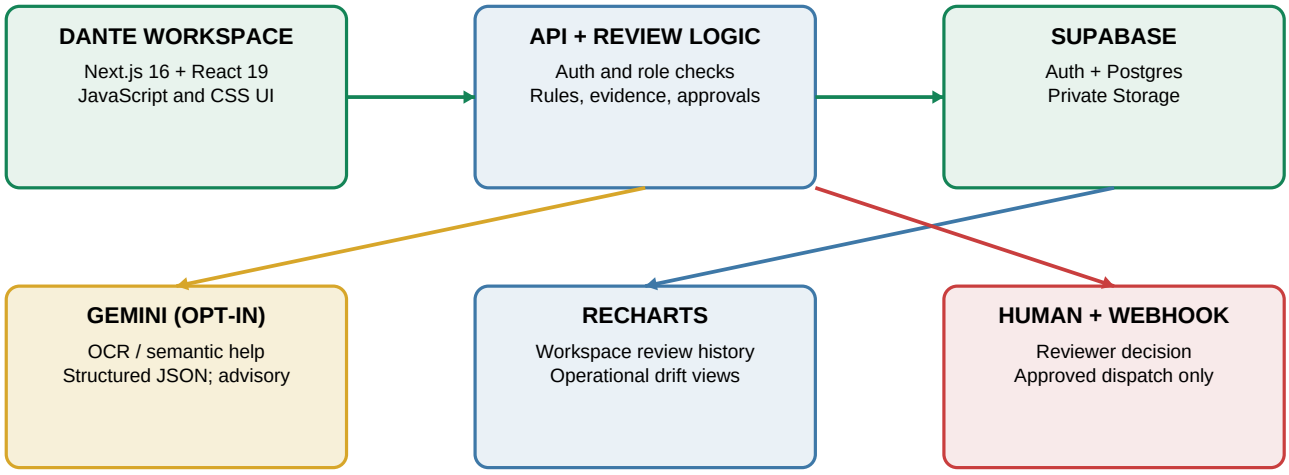
FOCUS AREA	STATUS	WHAT THE PROTOTYPE DOES TODAY
Connect real content sources	**PARTIAL**	Monitors public-page text on demand and records detected changes for review. A generic webhook can dispatch reviewer-approved content. Native CMS connectors, OAuth setup, and scheduled scans are not implemented.
Ingest policy documents with traceable evidence	**IMPLEMENTED**	Reads text PDFs and DOCX; preserves page or paragraph references; requires administrator approval. Scanned-page OCR is opt-in, page-cited, and flagged for original-document verification.
Measure and improve review accuracy	**PLANNED**	The current charts report review activity and content status. Dante does not yet have a human-labeled evaluation set or false-positive / missed-conflict metrics.

Improvements delivered in the current build

IMPROVEMENT	DELIVERED
Shared, authenticated workspaces	Supabase-backed state, roles, invitations, and private source storage.
Document and visual review	PDF/DOCX rule intake, consent-controlled OCR, image text review, and approved-logo reference comparison.
Review operations	Campaign/channel/team analytics, audit export, and a human-gated publishing webhook.
Safer language checks	Validated language tags and evidence quotes preserved in their original scripts; uncertain results remain for human review.
Loophole fixes	Persistent-workspace reset is blocked. Logo review now requires a valid approved image reference when selected.

Technical architecture

The browser uses the authenticated workspace API. Server routes verify the session and workspace membership before loading or saving workspace data. Optional AI calls run server-side after explicit consent.



LAYER	IMPLEMENTATION	ROLE
Application	Next.js 16.4.5, React 19, JavaScript, CSS	Authenticated shell, review UI, app routes, and workspace views.
API and rules	Next.js Route Handler plus server.mjs	Session / role checks, deterministic checks, Gemini orchestration, impact logic, and webhooks.
Data and identity	Supabase Auth and PostgreSQL	Workspace membership, sources, content, audit history, and saved state.
Files	Supabase private Storage	Original policy and logo files; admin-only short-lived document links.
AI	Gemini GenerateContent REST API	Opt-in structured OCR / semantic assistance; exact quotes are checked by the server.
Charts and deployment	Recharts; Vercel	Workspace analytics and target hosting. Actual deployment state must be smoke-tested.

Privacy boundary: Supabase and Gemini credentials remain server-side. The browser sends the user session to Dante; files are only forwarded to Gemini when the relevant consent is given.

Review reliability: deterministic checks handle exact facts and required terms. Gemini assists with semantic interpretation or extraction; the server validates evidence against the approved source, and unclear findings stay with a human reviewer.

Testing and reliability

CHECK	RESULT	SCOPE
Automated feature tests	**5 passed, 0 failed**	Exact fact conflict and evidence; OCR consent gate; OCR candidate citation and pending state; private original access by role; cross-language exact quotes; persistent reset protection.
Production build	**Passed**	Next.js compile, TypeScript phase, static route generation, and API route generation completed locally.

How to reproduce

```
node --test tests/round3-features.test.mjs
npm run build
```

Test limits

- Gemini is mocked in the automated tests. They verify request gating, parsing, evidence validation, and state transitions, but do not measure live-model accuracy.

- The OCR test uses a synthetic PDF with no text layer and a mocked transcript. A photographed/scanned guideline has not been evaluated against live Gemini OCR.

- No human-labeled accuracy set, precision/recall report, load test, native CMS integration test, scheduled-monitor test, or production Vercel smoke test has been completed.

- The page monitor reads server-delivered HTML on demand. JavaScript-rendered pages and scheduled polling are outside the current prototype.

Current safeguards

Human approval is required for extracted rules and publishing. Stale source versions block publishing approval. Workspace routes check authentication and role; original policy files stay private; OCR and image analysis are opt-in.

Product roadmap and future development

NEXT	MEASURABLE OUTCOME
Deploy and smoke-test the final build	Verify Supabase environment keys, OCR consent, PDF upload, role checks, original-file access, and the reviewer-gated webhook in the hosted environment.
Build an accuracy evaluation set	Label aligned, conflicting, and uncertain examples by language and channel; report precision, recall, false positives, and missed conflicts before tuning.
Connect real publishing systems	Add one CMS integration with scoped permissions, scheduled change checks, idempotent sync, and reviewer approval before guidance changes.
Improve multilingual and visual quality	Expand verified language coverage; compare OCR and semantic findings with human judgments; add measured contrast checks where technically reliable.
Scale team governance	Add reviewer assignments and notifications, configurable approval stages, retention controls, and richer audit exports.
Surface recurring issues	Group repeated findings and overrides with examples, then compare the patterns against labeled review outcomes.

Expected impact

Dante is designed to reduce time spent locating current guidance, surface stale claims before publishing, and give reviewers a traceable basis for decisions. Those benefits remain goals until measured with real teams and a labeled evaluation set.

Team

MEMBER	ROLE	CONTRIBUTION
Samriddha Chaudhury	Team Lead; Full Stack Developer	Product scope, backend, data model, source links, change-impact logic, review workflow, dashboards, and integration.
Md Rizwaan Rahaman	Tester	Demo content and testing.

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Dante keeps the decision with the people who own the brand.